Author Rebuttal for Submission 7872

We thank the reviewers for the positive assessment of object-level VAD attribution, the where–who–what formulation, and the ASP/TTP visual evidence design. We also thank the reviewers for suggestions on protocol transparency, fairness, sensitivity, runtime, and wording. Below we clarify under-specified points and list concrete revisions without changing the core method.

Dataset Scope and Query Fairness

4AFu, JWmb, pkAN, and tDMF raised dataset coverage, benchmark construction, and query-prior concerns. We clarify that O-ShT/O-Ave are object-centric attribution layers for who/what evaluation, and each query is evaluated over all test videos rather than known-class positives.

O-ShT/O-Ave are built on ShanghaiTech/Avenue because these surveillance datasets contain many pedestrian-centered anomalies and object interactions, making them suitable for testing whether zero-shot VAD can identify *what* happens and *which visible entity* supports the decision. O-ShT contains 11 event queries, 107 annotated videos, 91 positive video-query pairs, 127 temporal segments, and 143 object annotations. O-Ave contains 4 queries, 21 annotated videos, 17 positive pairs, 47 segments, and 47 object annotations. Each positive pair has an anomalous segment, representative frame, entity box, and object/action description.

The setting is query-conditioned, but not oracle class-only: each query is evaluated against all test videos, including normal and non-target abnormal videos, so false positives and false negatives are both penalized. We will include the query list, per-query statistics, annotation rules, and ambiguity handling. UCF-Crime/XD-Violence are important for generic VAD, while many categories emphasize video-level events such as explosions, shooting, robbery, or fighting; their public annotations are primarily video/temporal labels rather than event-query aligned object boxes. We will therefore use them for detection-only comparisons where appropriate and describe object-level extension as future annotation.

All baselines will use the same query vocabulary and AUC protocol. Methods without native boxes will be compared on AUC and semantic/event accuracy rather than forced into bbox-based O-Acc; any shared detector-proposal adapter will be marked as an adapter, not native output.

Metrics, BBoxes, and Thresholds

4AFu, odWX, pkAN, and tDMF asked how frame scores, object boxes, IoU, and Sentence-BERT thresholds are defined. We will make score construction and object attribution diagnostics explicit.

Branch scores are aligned with frame labels, concatenated over the test set, Gaussian-smoothed, min–max normalized, and evaluated by micro ROC/AUC. We will also report video-level AUC using max and top-10% mean aggregation. The predicted box is the CLIP-ranked ASP detector proposal, not a box recovered from the soft mask.

To make O-Acc auditable, we will separate semantic object/action matching for explanations from localization accuracy by proposal–GT IoU. We will report Loc/O-Acc@0.3 and @0.5: @0.3 measures coarse attribution for small/occluded surveillance actors, while @0.5 is the stricter reference. Reporting both thresholds makes the metric transparent across coarse attribution and stricter localization. Sentence-BERT 0.75 was chosen to tolerate paraphrases while rejecting category confusions; we will add 0.70/0.75/0.80 sensitivity and confusion examples.

Ablation, Sensitivity, and Efficiency

pkAN, JWmb, 4AFu, and odWX requested fuller ablations, sensitivity analysis, video AUC, and runtime reporting. Using saved branch scores under the same Qwen3-VL backbone, we provide compact diagnostics showing branch complementarity.

Setting	ShT-F	ShT-V	Ave-F	Ave-V
GCP only	0.9613	0.9794	0.8483	0.8674
ASP only	0.9044	0.9621	0.7881	0.8463
TTP only	0.8447	0.8615	0.8278	0.8750
Full	0.9849	0.9930	0.8510	0.8712
w/o GCP	0.9522	0.9830	0.8338	0.8750
w/o ASP	0.9702	0.9799	0.8529	0.8636
w/o TTP	0.9846	0.9930	0.8447	0.8674

GCP is strongest on average for detection, while ASP/TTP add object and motion evidence and help car, loitering, throwing, and running. We will add CoVE-off and uniform-anchor diagnostics. Weight sensitivity is modest: ShT full/default, paper-balanced, and equal fusion obtain 0.9849/0.9846/0.9844; Avenue obtains 0.8510/0.8518/0.8515. Since the pipeline is frozen and deterministic, bootstrap confidence intervals over videos/queries are more informative than multi-seed variance.

For scalability, we will report call counts, cacheability, and hardware-dependent latency in the revision, while emphasizing the scaling structure. O-VAD is inference-only with no task-specific training or fine-tuning; it uses cached detection/tracking once per video and one batched LVLM call per processed clip with three prompt instances. After caching visual proposals, the dominant LVLM cost scales linearly with event queries, and query pruning or shared proposal reuse can reduce calls for larger vocabularies.

Novelty and Relation to Prior Work

JWmb, pkAN, and tDMF asked us to distinguish O-VAD from system integration and recent LVLM-based VAD/explanation work. O-VAD's contribution lies in structured evidence construction for zero-shot attribution, rather than relying on a new trainable detector.

GCP keeps scene context, ASP focuses the likely anomalous subject without cropping away the scene, TTP turns track-level motion into a static LVLM-consumable prompt, and CoVE fuses branch evidence into auditable where/who/what outputs. This differs from raw LVLM prompting because the model receives explicit spatial and temporal evidence before attribution. We will expand discussion of LAVAD, HAWK, Sherlock, VANGUARD, and role-aware/instance-aligned grounding.

For the comparison to LAVAD, the central claim is the effect of explicit evidence construction under the same backbone and query protocol: it improves interpretability and exposes where attribution succeeds or fails. O-VAD is complementary to supervised/weakly supervised grounding pipelines.

Claim Revision and Failure Analysis

4AFu, pkAN, and tDMF asked how upstream detector/tracker errors and causal/non-destructive wording should be handled. We will clarify these points and add failure analysis to make the pipeline behavior auditable.

The reported O-Acc values complement AUC by measuring whether the anomalous entity is correctly attributed, not only whether an event is detected. We will add failure cases separating detector misses, wrong ASP proposal selection, tracker switches, trajectory artifacts, and LVLM semantic errors, turning the modular pipeline into an auditable error-tracing process.

We also appreciate the wording suggestions. "Without information loss" will be revised to "context-preserving" or "less destructive than cropping," since ASP smoothing and TTP overlays modify pixels. "Causal reasoning" will be revised to "structured evidence-based attribution," which better matches the evidence-driven reasoning performed by the framework. These changes clarify the paper's scope while preserving its main contribution.